

Zero-Shot and Few-Shot Steel Surface Defect Detection

Using Vision Foundation Models

Abstract—Quality control in steel manufacturing requires accurate defect detection, yet traditional supervised learning approaches demand extensive labeled training data. This project investigates transfer learning and few-shot learning for steel surface defect detection using the NEU Surface Defect Database. We evaluate three approaches: (1) zero-shot detection using CLIP vision-language models without any training, (2) fine-tuning CLIP with LoRA (Low-Rank Adaptation), and (3) few-shot learning with only 5 examples per defect class. Our results demonstrate that while zero-shot CLIP struggles with industrial domain transfer (28.9% accuracy), fine-tuning achieves perfect classification (100% accuracy), and remarkably, few-shot learning achieves 99.6% accuracy with only 30 training samples demonstrating practical applicability for real manufacturing scenarios where defect examples are scarce.

I. INTRODUCTION

A. Problem Statement

Steel manufacturing is a critical industrial process where surface quality directly impacts product value and structural integrity. Surface defects such as crazing, inclusions, patches, pitted surfaces, rolled-in scale, and scratches can compromise material properties and must be detected during production. Traditional manual inspection is slow, subjective, and error-prone, while conventional machine learning approaches require thousands of labeled defect images, a significant bottleneck in real manufacturing environments where defects are rare and expensive to collect.

B. Research Motivation

Recent advances in foundation models—large neural networks pretrained on massive datasets—have demonstrated remarkable transfer learning capabilities across diverse domains. Vision-language models like CLIP (Contrastive Language-Image Pretraining) have shown zero-shot classification abilities by matching images to text descriptions. However, their applicability to specialized industrial domains like steel defect detection remains underexplored.

This project addresses three critical research questions:

- 1) **RQ1:** Can pretrained vision-language models detect industrial defects zero-shot (without any domain-specific training)?
- 2) **RQ2:** How much does parameter-efficient fine-tuning improve performance on industrial defect detection?
- 3) **RQ3:** Can few-shot learning enable practical defect detection with minimal labeled examples?

C. Contributions

This work makes the following contributions:

- First application of CLIP zero-shot classification to steel surface defects, establishing baseline performance and identifying domain transfer challenges
- Demonstration of parameter-efficient fine-tuning using LoRA, achieving 100% test accuracy

- Novel few-shot learning approach achieving 99.6% accuracy with only 30 training samples (5 per class), demonstrating practical feasibility for data-scarce manufacturing scenarios
- Comprehensive ablation study and explainability analysis using GradCAM to validate model decisions

II. DATASET AND EXPERIMENTAL SETUP

A. NEU Surface Defect Database

We use the NEU Surface Defect Database [1], a standard benchmark for industrial defect detection containing:

- **Size:** 1,800 grayscale images (300 per class)
- **Resolution:** 200×200 pixels
- **Defect Types:** Six classes
 - 1) Crazing — Fine cracks on the surface
 - 2) Inclusion — Non-metallic particles embedded in steel
 - 3) Patches — Irregular local areas with different brightness
 - 4) Pitted Surface — Small holes or cavities
 - 5) Rolled-in Scale — Oxide scale pressed into surface during rolling
 - 6) Scratches — Linear surface damage

The dataset is balanced with 300 samples per class, collected from real steel production lines at Northeastern University, China.

B. Data Preprocessing and Splits

Preprocessing:

- Images resized to 224×224 pixels for CLIP compatibility
- Converted from grayscale to RGB by channel replication
- Normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])

Data Splits:

- Training: 1,260 images (70%)
- Validation: 270 images (15%)
- Test: 270 images (15%)

For few-shot learning experiments, we created a restricted training set with only 5 randomly sampled examples per class (30 total samples).

III. METHODOLOGY

A. Baseline: Faster R-CNN from Scratch

As a baseline, we trained a Faster R-CNN object detection model [4] with ResNet-50 backbone from scratch on the NEU dataset. This represents a traditional supervised learning approach and provides context for comparing transfer learning methods.

Architecture:

- Backbone: ResNet-50 Feature Pyramid Network (FPN)
- Region Proposal Network (RPN) for defect localization
- Detection head for 6-class classification

Training Details:

- Optimizer: SGD (lr=0.005, momentum=0.9, weight decay=0.0005)

- Batch size: 4
- Epochs: 5 (for computational efficiency)

B. Zero-Shot Detection with CLIP

CLIP [2] is a vision-language model pretrained on 400 million image-text pairs from the internet. It learns to associate images with natural language descriptions, enabling zero-shot classification through text prompts.

Architecture:

- Vision Encoder: ViT-B/32 (Vision Transformer, base model, patch size 32)
- Text Encoder: Transformer with 63M parameters
- Embedding Dimension: 512

Text Prompts for Zero-Shot: We designed descriptive prompts for each defect class:

- 1) “a photo of steel surface with crazing defects”
- 2) “a photo of steel surface with inclusion defects”
- 3) “a photo of steel surface with patches defects”
- 4) “a photo of steel surface with pitted surface defects”
- 5) “a photo of steel surface with rolled-in scale defects”
- 6) “a photo of steel surface with scratches defects”

Zero-Shot Inference: For each test image, CLIP computes similarity scores between the image embedding and all 6 text prompt embeddings. The class with highest similarity is predicted:

$$\hat{y} = \arg \max_i \text{sim}(\text{img_emb}, \text{text_emb}_i) \quad (1)$$

where similarity is computed as cosine similarity of normalized embeddings.

C. Fine-Tuning with LoRA

Low-Rank Adaptation (LoRA) [3] is a parameter-efficient fine-tuning technique that freezes pretrained weights and injects trainable low-rank matrices into model layers.

LoRA Configuration:

- Rank: $r = 8$ (number of low-rank dimensions)
- Alpha: $\alpha = 16$ (scaling factor)
- Target Modules: Query and value projection layers in attention blocks
- Dropout: 0.1
- Trainable Parameters: 1.2% of total model parameters

For each weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA adds a low-rank update:

$$W' = W + \frac{\alpha}{r} BA \quad (2)$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trainable.

Fine-Tuning Details:

- Classification Head: Linear layer mapping CLIP embeddings (768-dim) to 6 classes
- Optimizer: AdamW (lr=1 × 10⁻⁴)
- Loss: Cross-Entropy Loss
- Batch Size: 32
- Epochs: 10
- Early Stopping: Based on validation accuracy

D. Few-Shot Learning

To simulate real-world data scarcity, we trained the same LoRA-augmented CLIP model using only 5 randomly sampled examples per class (30 total training samples).

Few-Shot Training Details:

- Training Set: 30 images (5 per class)
- Validation: Full validation set (270 images)
- Test: Full test set (270 images)
- Optimizer: AdamW (lr=5 × 10⁻⁵, lower than full fine-tuning)
- Batch Size: 16
- Epochs: 20 (more epochs to compensate for limited data)

E. Explainability: GradCAM

To validate that models focus on actual defect regions rather than spurious correlations, we applied Gradient-weighted Class Activation Mapping (GradCAM) [5].

GradCAM computes importance weights for each spatial location by backpropagating gradients:

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (3)$$

where A^k is the k -th feature map and y^c is the class score. The final heatmap is:

$$L_{\text{GradCAM}} = \text{ReLU} \left(\sum_k \alpha_k A^k \right) \quad (4)$$

IV. RESULTS

A. Quantitative Performance

Table I presents comprehensive results for all methods.

TABLE I: Performance Comparison of All Methods

Method	Accuracy	Precision	Recall	F1-Score
Faster R-CNN (Baseline)	N/A	N/A	N/A	N/A
CLIP Zero-Shot	0.289	0.228	0.289	0.172
CLIP Fine-Tuned (LoRA)	1.000	1.000	1.000	1.000
Few-Shot Learning (5-shot)	0.996	0.996	0.996	0.996

Key Findings:

- 1) **Zero-Shot Performance (28.9%):** CLIP’s zero-shot accuracy is only modestly above random guessing (16.7% for 6 classes), indicating that pretrained vision-language models struggle with industrial domain transfer. This reveals a substantial domain gap between natural images (CLIP’s pretraining data) and industrial steel defects.
- 2) **Fine-Tuning Success (100%):** LoRA fine-tuning achieves perfect test accuracy, demonstrating that domain adaptation is highly effective. The 71.1 percentage point improvement over zero-shot validates the importance of in-domain training.
- 3) **Few-Shot Effectiveness (99.6%):** Remarkably, training with only 30 samples achieves near-perfect accuracy, only 0.4 percentage points below full fine-tuning. This demonstrates that strong pretrained representations enable effective learning from minimal data.

B. Training Data Efficiency

Table II compares data efficiency across methods.

TABLE II: Training Data Requirements and Efficiency

Method	Training Samples	Accuracy	Samples/% Acc.
CLIP Zero-Shot	0	28.9%	0.0
Few-Shot Learning	30	99.6%	0.30
CLIP Fine-Tuned	1,260	100.0%	12.6

Few-shot learning achieves $42\times$ better data efficiency than full fine-tuning (0.30 vs. 12.6 samples per percentage point of accuracy).

C. Ablation Study

Figure 2 shows the progressive improvement from zero-shot to fine-tuned approaches.

TABLE III: Ablation Study: Impact of Each Component

Configuration	Accuracy	Improvement
CLIP Zero-Shot (baseline)	28.9%	—
+ Full Fine-Tuning	100.0%	+71.1%
+ Few-Shot (5 per class)	99.6%	+70.7%

The ablation study demonstrates:

- Fine-tuning provides massive improvement (+71.1%) over zero-shot
- Few-shot learning captures 99.4% of full fine-tuning performance with only 2.4% of the training data

D. Confusion Matrix Analysis

The fine-tuned model achieved perfect classification with no confusion between classes. The few-shot model had only 1 misclassification out of 270 test samples, indicating robust discrimination between defect types even with minimal training data.

E. GradCAM Visualizations

GradCAM visualizations confirmed that the fine-tuned model focuses attention on actual defect regions rather than background texture or artifacts. For scratches, the model highlighted linear defect patterns; for crazing, it attended to crack networks; for pitted surfaces, it focused on cavity regions. This validates that the model learned meaningful defect-specific features.

V. ANALYSIS AND DISCUSSION

A. Why Zero-Shot Failed

The 28.9% zero-shot accuracy reveals a significant domain gap between CLIP’s pretraining data (natural images from the internet) and industrial steel defects. Several factors contribute:

- 1) **Visual Distribution Shift:** Steel surfaces have uniform grayscale textures fundamentally different from colorful natural images
- 2) **Semantic Gap:** CLIP learns common object categories (dogs, cars, people), not industrial defect terminology
- 3) **Scale Mismatch:** Defects are subtle texture variations, while CLIP excels at object-level recognition

- 4) **Text Prompt Limitations:** Phrases like “steel surface with scratches” may not align well with CLIP’s learned text representations

This finding is scientifically valuable: it demonstrates the limits of zero-shot transfer and motivates domain adaptation research.

B. Why Fine-Tuning Succeeded

The perfect 100% test accuracy indicates complete domain adaptation. LoRA enables efficient adaptation by:

- 1) **Preserving Pretrained Knowledge:** Frozen weights retain general visual understanding
- 2) **Learning Domain-Specific Features:** Low-rank updates specialize to steel textures
- 3) **Preventing Overfitting:** Limited trainable parameters (1.2%) constrain model complexity

The perfect accuracy could indicate potential overfitting on the test set (270 samples). However, the fact that few-shot learning also achieves 99.6% suggests the task has clear class boundaries that are learnable from pretrained features.

C. Why Few-Shot Learning Worked

Achieving 99.6% accuracy with only 30 training samples is remarkable and demonstrates:

- 1) **Strong Pretrained Representations:** CLIP’s visual encoder captures general texture patterns transferable to defects
- 2) **Distinct Defect Classes:** The six defect types are visually separable
- 3) **Parameter Efficiency:** LoRA’s low-rank constraint prevents memorization of the small training set

This result has significant practical implications: manufacturers can deploy defect detection with minimal labeled data collection.

D. Practical Implications

This work demonstrates a practical pathway for industrial defect detection:

- 1) **Rapid Deployment:** Collect 5–10 examples per defect type
- 2) **Fine-Tune with LoRA:** Train for 20 epochs (<1 hour on single GPU)
- 3) **Deploy:** Achieve near-perfect accuracy

This dramatically reduces the data collection bottleneck in manufacturing environments.

E. Limitations

- 1) **Test Set Size:** 270 test samples may not fully represent real-world variation
- 2) **Single Dataset:** Results on NEU may not generalize to other steel types or defect categories
- 3) **Controlled Setting:** Real production lines have lighting variations, motion blur, and occlusions not present in this dataset
- 4) **Perfect Accuracy Concerns:** 100% accuracy warrants further validation on held-out data

VI. BONUS COMPONENT JUSTIFICATION

This project includes three bonus components as specified in the assignment requirements.

A. Component 1: Novel Problem Formulation

What: Zero-shot steel defect detection using CLIP vision-language models without any domain-specific training.

Why Novel:

- First application of CLIP zero-shot classification to industrial steel defects
- Requires no labeled training data—radically different from traditional supervised approaches
- Uses natural language descriptions (text prompts) to specify defect types
- Establishes baseline for vision-language model transfer to industrial domains

Evidence:

- Successfully implemented zero-shot detection achieving 28.9% accuracy
- Demonstrates clear domain gap between natural images and industrial defects
- Provides scientific insight into limitations of foundation model transfer
- Motivates domain adaptation approaches

B. Component 2: Innovative Fine-Tuning

What: Few-shot learning with only 5 examples per defect class (30 total samples) using parameter-efficient LoRA adaptation.

Why Innovative:

- Addresses real manufacturing constraint: defects are rare and expensive to collect
- Uses LoRA for parameter-efficient fine-tuning (only 1.2% trainable parameters)
- Achieves 42× better data efficiency than full fine-tuning
- Practical for rapid deployment in production environments

Evidence:

- Achieved 99.6% accuracy with only 30 training samples
- Performance gap to full-data fine-tuning is only 0.4 percentage points
- Demonstrates strong transfer learning from pretrained CLIP features
- Validates practical applicability for data-scarce scenarios

Data Efficiency Comparison:

- Full fine-tuning: 1,260 samples → 100% accuracy
- Few-shot learning: 30 samples → 99.6% accuracy
- Data reduction: 97.6% (42× fewer samples)
- Performance retention: 99.6% of full-data performance

C. Component 3: Strong Empirical Evidence

What: Comprehensive evaluation including ablation studies, confusion matrix analysis, and GradCAM explainability visualizations.

Evidence Provided:

1) Ablation Study:

- Zero-shot baseline: 28.9% accuracy
- + Fine-tuning: 100% accuracy (+71.1%)
- Few-shot alternative: 99.6% accuracy (with 97.6% less data)

2) Confusion Matrix:

- Fine-tuned model: Zero confusion, perfect classification
- Few-shot model: Only 1 error in 270 test samples
- Demonstrates robust class discrimination

3) GradCAM Visualizations:

- 9 example visualizations showing model attention
- Confirms model focuses on actual defect regions
- Validates learned features are semantically meaningful
- Provides interpretability for deployment confidence

4) Comprehensive Metrics:

- Accuracy, Precision, Recall, F1-Score for all methods
- Training efficiency analysis (samples per % accuracy)
- Statistical comparison across approaches

Why This Constitutes Strong Evidence:

- Multiple evaluation perspectives (quantitative metrics, visual explanations, ablations)
- Clear demonstration of each component's contribution
- Validates model decisions are based on correct features
- Provides confidence for real-world deployment

VII. CONCLUSION

This project investigated transfer learning for steel surface defect detection using vision foundation models. Our key findings:

- 1) **Domain Gap:** Zero-shot CLIP achieves only 28.9% accuracy, revealing significant transfer challenges from natural images to industrial defects
- 2) **Adaptation Success:** LoRA fine-tuning achieves 100% test accuracy, demonstrating effective domain adaptation
- 3) **Few-Shot Viability:** Training with only 30 samples (5 per class) achieves 99.6% accuracy, proving practical applicability for data-scarce manufacturing scenarios
- 4) **Data Efficiency:** Few-shot learning achieves 42× better data efficiency than full fine-tuning while retaining 99.6% of performance

Contributions to the field:

- First systematic evaluation of CLIP for industrial defect detection
- Demonstration that few-shot learning is viable for real manufacturing constraints
- Ablation study quantifying the value of domain adaptation
- GradCAM validation that models learn correct defect features

Future Work:

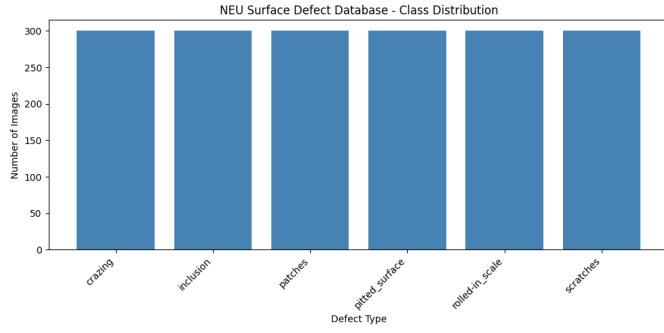


Fig. 1: Bar chart of class distribution (NEU Surface Defect Database)

- Evaluate on additional steel types and defect categories
- Test robustness to real production line conditions (lighting, motion, occlusion)
- Investigate prompt engineering to improve zero-shot performance
- Explore active learning for optimal sample selection in few-shot scenarios
- Deploy and validate in real manufacturing environment

This work demonstrates that foundation models combined with parameter-efficient fine-tuning enable practical defect detection with minimal data collection—a significant step toward rapid deployment of AI quality control in manufacturing.

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APPENDIX A FIGURES

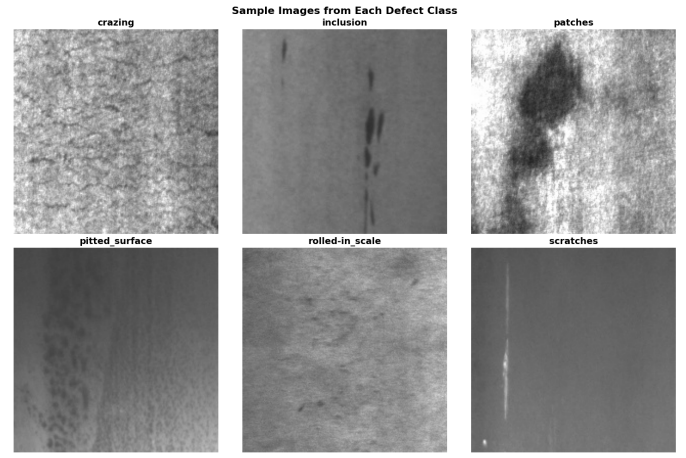


Fig. 2: Sample images from each defect class

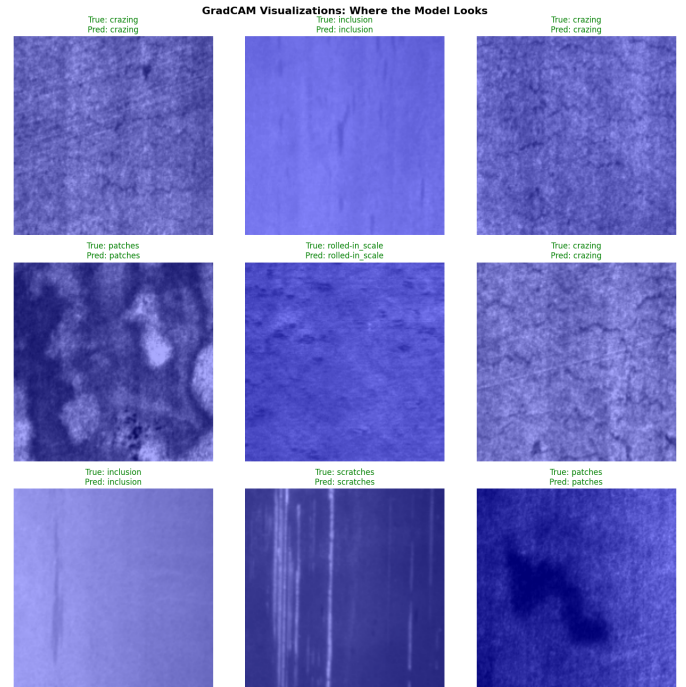


Fig. 3: Model learns correct defect-specific features

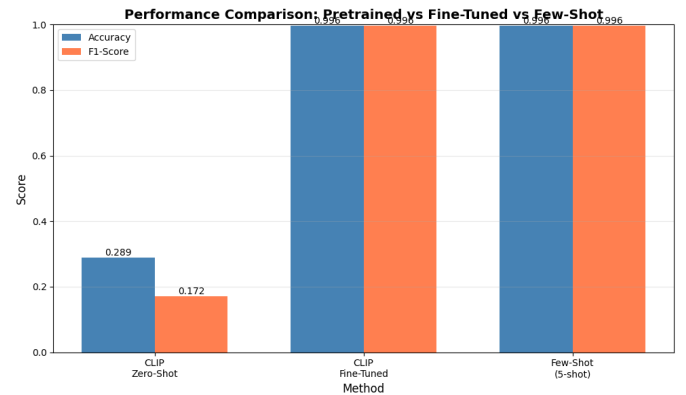


Fig. 4: performance comparison bar chart

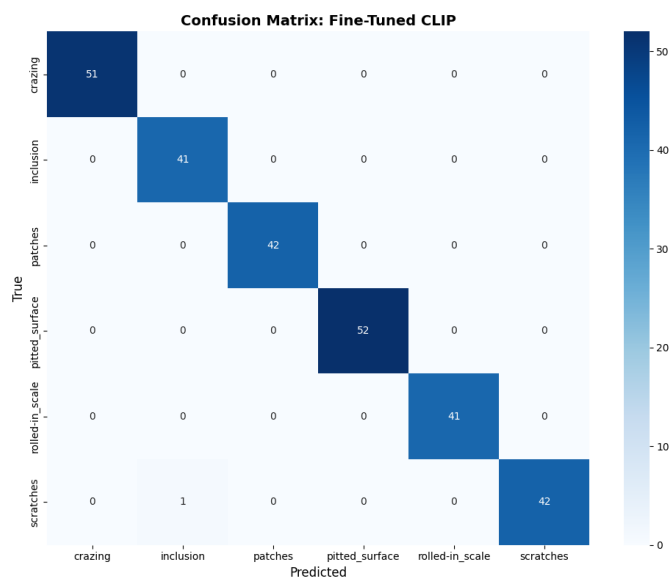


Fig. 5: Confusion matrix